CSE 102 Assignment 4 Report

Assignment Objective:

The purpose of this assignment is to simulate a rocket's vertical flight using physics-based mathematical modeling. The flight is described by a quadratic equation considering the effects of gravity, launch velocity, and height. The simulation provides a text-based graphical representation and saves trajectory data for further analysis.

enter_rocket_parameters()

What it does:

This function asks the user to enter three things:

- Initial velocity (how fast the rocket goes up),
- Gravity (how strong the pull of gravity is),
- Launch height (how high the rocket starts from the ground).

Then it saves these values into a file called rocket_data.txt.

```c
void enter_rocket_parameters() {
    FILE *dosya = fopen("rocket_data.txt", "w");
    if (dosya == NULL) {
        printf("Error opening file.\n");
        return;
    }

    double velocity, gravity, launch_height;

    // Get parameters from the user
    printf("Enter initial velocity (m/s): ");
    scanf("%lf", &velocity);

    printf("Enter gravity (m/s², default 9.8): ");
    scanf("%lf", &gravity);

    printf("Enter launch height (m): ");
    scanf("%lf", &launch_height);

    // Save parameters to file
    fprintf(dosya, "%lf %lf %lf", velocity, gravity, launch_height);
    fclose(dosya);

    printf("Rocket parameters saved to rocket_data.txt!\n");
}
```

draw_graph(double velocity, double gravity, double launch_height)

What it does:

It calculates the rocket's height at different time points and draws a text-based graph using #.

```c
void draw_graph(double velocity, double gravity, double launch_height) {
    printf("Reading rocket parameters from file...\n");
    printf("Equation: h(t) = -%.2lf * t^2 + %.2lf * t + %.2lf\n", 0.5 * gravity, velocity
    printf("Simulating trajectory...\n");
```

save_trajectory_to_file(double velocity, double gravity, double launch_height)

What it does:

It writes the graph and some flight data (like maximum altitude and flight time) into a file called trajectory.txt.

```c
void save_trajectory_to_file(double velocity, double gravity, double launch_height) {
    FILE *file = fopen("trajectory.txt", "w");

    // Calculate trajectory info
    double hmax = (velocity * velocity) / (2 * gravity) + launch_height;
    double t_peak = velocity / gravity;
    double t_flight = (velocity + sqrt(velocity * velocity + 2 * gravity * launch_height)) / gravity;

    // Write flight data to file
    fprintf(file, "Maximum Altitude: %.2lf meters\n", hmax);
    fprintf(file, "Total Flight Duration: %.2lf seconds\n\n", t_flight);
```

main()

What it does:

Displays the main menu and allows the user to choose an action. It loops until the user selects "Exit".

```c
printf("1. Enter rocket launch parameters\n");
printf("2. Simulate and graph the rocket's trajectory\n");
printf("3. Display and save the trajectory data\n");
printf("4. Exit the program\n");
```

OUTPUTS

```
1. Enter rocket launch parameters
2. Simulate and graph the rocket's trajectory
3. Display and save the trajectory data
4. Exit the program
Select an option: 1
Enter initial velocity (m/s): 50
Enter gravity (m/s², default 9.8): 9.8
Enter launch height (m): 10
Rocket parameters saved to rocket_data.txt!
```

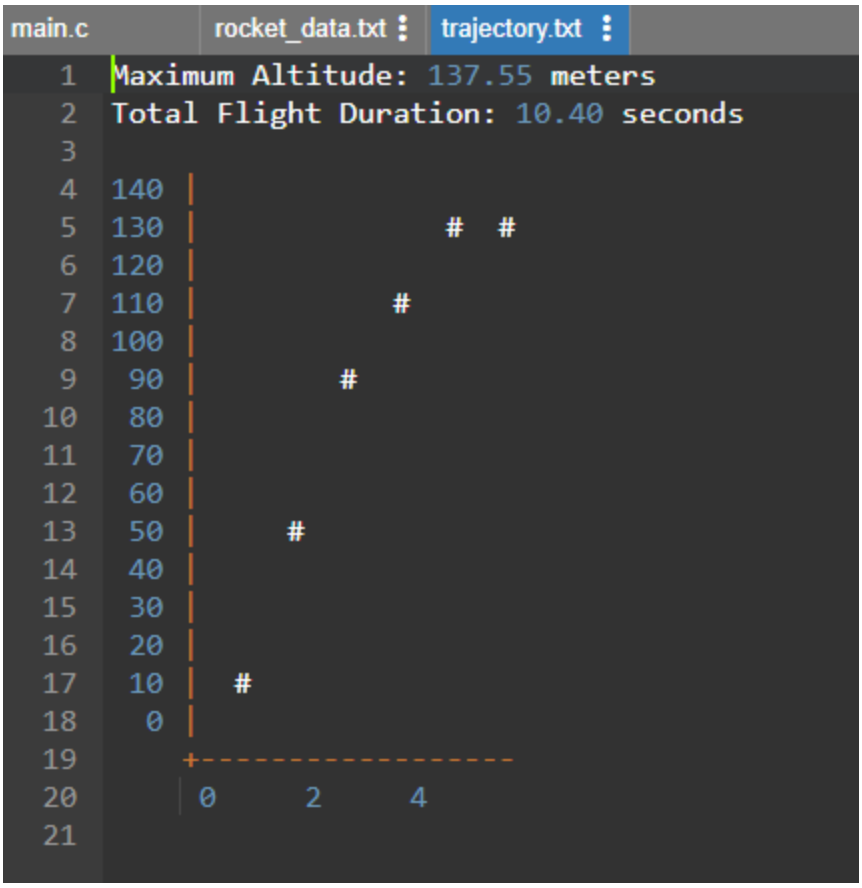
```
1. Enter rocket launch parameters
2. Simulate and graph the rocket's trajectory
3. Display and save the trajectory data
4. Exit the program
Select an option: 2
Reading rocket parameters from file...
Equation: h(t) = -4.90 * t^2 + 50.00 * t + 10.00
Simulating trajectory...
140 |
130 |               #   #
120 |
110 |            #
100 |
 90 |         #
 80 |
 70 |
 60 |
 50 |      #
 40 |
 30 |
 20 |
 10 |   #
  0 |
    +------------------
     0      2      4
```

```
1. Enter rocket launch parameters
2. Simulate and graph the rocket's trajectory
3. Display and save the trajectory data
4. Exit the program
Select an option: 3
Saving trajectory data...

Maximum Altitude: 137.55 meters
Total Flight Duration: 10.40 seconds
Graph saved to trajectory.txt!

1. Enter rocket launch parameters
2. Simulate and graph the rocket's trajectory
3. Display and save the trajectory data
4. Exit the program
Select an option: 4


...Program finished with exit code 0
Press ENTER to exit console.
```



Conclusion

This project provides a simplified simulation of rocket motion using classical physics principles and ASCII graphing. It successfully meets all criteria set by the assignment and is extensible for more complex simulations.